Industrial Processing and Operations

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Industrial Processing and Operations (A34181)

Short Title:	Industrial Processing and Ops		
Faculty	Science and Computing		
Department	Science		
Cluster	Pharmaceutical Science		
Author	KMURPHY		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

This module describes the overall operation, unit processes, equipment and documentation utilised in the modern pharmaceutical and biopharmaceutical industries. The module will also focus on key elements in pharmaceutical scale-up as part of a drug development programme. The student will be provided with an understanding of Quality by Design principles, and the various tools used to identify and reduce product variability. The course is designed to be suitable for both pharmaceutical and biopharmaceutical students.

Programmes

`SCIE-0066`	BSc in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_D`)	3 / 6 / M
`SCIE-0066`	BSc in Pharmaceutical Science (`WD_SPHSC_D`)	3 / 6 / M

Indicative Content

- Pharmaceutical operations: facility design, layout, process flow, process control, process safety; green chemistry
- Pharmaceutical unit operations and equipment: reactors, reactor kinetics, calorimetry, crystallisation, filtration, drying, powder properties
- Pharmaceutical unit operations and equipment: separation equipment, distillation, solvent recovery
- Biopharmaceutical unit operations and equipment: fermentation, bioreactor design, scale-up, harvesting, filtration, purification
- Biopharmaceutical unit operations and equipment: filling, lyophilisation, packaging
- Pharmaceutical/biopharmaceutical processing documentation: BMRs, BPRs, Out of Specification results, labelling
- Aspects in scale-up and pilot plant production
- Quality by Design: critical quality attributes, risk assessment, experimental design, Process Analytical Technology (PAT)

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Demonstrate the importance of design, layout and process control to pharmaceutical operations.
2. Illustrate and explain the operation and function of pharmaceutical and biopharmaceutical processing equipment.
3. Compare individual processes and their underlying theory and describe associated documentation.
4. Explore and categorise milestones and hazards in pharmaceutical scale-up.
5. Interpret and classify risk and learn how to reduce risk through process understanding and experimental design.

Learning and Teaching Methods

- Lectures.

- Field trips.
- Group work.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	20%	3
Group Project	10%	5
In-Class Assessment	70%	1,2,3,4

Assessment Criteria

<40%: The student will have failed to attain the learning outcomes at even a basic level.
40%-49%: The student will have attained the learning outcomes at a basic level.
50%-59%: The student will be able to demonstrate understanding of some of the complexities of the topics.
60%-69%: The student will demonstrate a more detailed knowledge of all of the topics covered, and will have the ability to interpret data.
70%-100%: The student will demonstrate a comprehensive understanding of all of the material covered, including calculations.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Independent Learning	99	

Essential Material

- Mollah, A.H., M. Long and M. Baseman. _Risk Management Applications in Pharmaceutical and Biopharmaceutical Applications_. USA: Wiley, 2013.
- Nuslim, S. _Active Pharmaceutical Ingredient. Development, Manufacture and Regulation_. USA: Taylor and Francis, 2005.
- Walsh, G. _Biopharmaceuticals : biochemistry and biotechnology_. Chichester: Wiley, 1998.

Supplementary Material

- Cox, S. _Pharmaceutical Manufacturing Handbook: Production and Processes_. USA: Wiley, 2008.
- Doble, M. and A.K. Krutheventi. _Green Chemistry and Processes_. London: Academic Press, 2007.
- McCabe, W., J.C. Smith and P. Harriott. _Unit operations of chemical engineering_. London: McGraw-Hill, 1993.
- Moulijn, J.A., I.M. Makkee and A.E. Van Diepen. _Chemical process technology_. New York: John Wiley & Sons, 2001.